A Quick Tutorial on Using LaTeX with Overleaf

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1 Overview

This tutorial provides a practical, beginner-friendly introduction to writing documents in LaTeX using Overleaf. By the end, you will be able to create a project, write equations, insert figures, manage references, and export a PDF.

2 What are LaTeX and Overleaf?

- **LaTeX** is a typesetting system widely used for technical documents, especially those containing mathematics, figures, and structured formatting.
- **Overleaf** (https://www.overleaf.com) is a web-based LaTeX editor that compiles your source into a PDF and supports collaboration (similar to Google Docs).

3 Create an Overleaf Account and Start a Project

1. Go to https://www.overleaf.com and create an account using your UMBC email address. The College of Engineering and Information Technology at the University of Maryland, Baltimore County, provides Overleaf Professional features to its students, faculty, and staff.
2. Click **New Project** → **Blank Project**.
3. Name your project (e.g., `My_First_LaTeX_Doc`) and click **Create**.
4. Open `main.tex` (or `<projectname>.tex`) and edit the text.
5. Click **Recompile** to build the PDF.

3.1 Minimal Working Example (MWE): LaTeX Document

Copy and paste the following into your `main.tex` file.

```
\documentclass[12pt]{article}

\usepackage{amsmath}
\usepackage{amssymb}
\usepackage{graphicx}
\usepackage{geometry}
\geometry{margin=1in}

\title{My First LaTeX Document}
\author{Your Name}
\date{\today}

\begin{document}
\maketitle

\section{Introduction}
This is a simple LaTeX document.

\end{document}
```

Click **Recompile** to build the PDF.

4 Basic Document Structure

A typical LaTeX file has two main parts.

- **Preamble:** everything *before* `\begin{document}`. This is where you choose the document type (class), load packages, define custom commands, set margins, and configure formatting.
- **Document body:** everything between `\begin{document}` and `\end{document}`. This is where the content of your document goes (text, figures, equations, tables, etc.).

4.1 Sections, Subsections, and Subsubsections

LaTeX provides a built-in hierarchy of headings to help you organize content. The most common are

- `\section{...}`: top-level divisions (major topics).
- `\subsection{...}`: subdivisions within a section.
- `\subsubsection{...}`: finer divisions within a subsection.

Why use heading structure? Using headings improves readability, makes your writing easier to navigate, and enables LaTeX features such as automatic numbering, consistent formatting, and automatic table of contents generation.

A common convention. Do not overuse `\subsubsection`. For most documents, `\section` and `\subsection` are sufficient. Use `\subsubsection` when the structure genuinely helps the reader.

4.2 Minimal Working Example: Heading Structure

The following MWE demonstrates how sections, subsections, and subsubsections appear, and how they can automatically generate a table of contents.

```latex
\documentclass[11pt]{article}

\usepackage{geometry}
\geometry{margin=1in}

\title{Heading Structure Example}
\author{Your Name}
\date{\today}

\begin{document}
\maketitle

\tableofcontents
```

```latex
\bigskip

\section{Introduction}
This section introduces the document and outlines the main goals.

\subsection{Motivation}
This subsection explains why the topic matters and provides context.

\subsubsection{A Specific Motivation}
This subsubsection highlights one specific reason and adds detail.

\subsection{Contributions}
This subsection lists key contributions.
\begin{itemize}
  \item Contribution 1
  \item Contribution 2
\end{itemize}

\section{Methods}
This section explains the approach used.

\subsection{Model}
Describe the model assumptions and setup.

\subsection{Algorithm}
Describe the algorithmic steps clearly.

\section{Results}
Present results with figures, tables, and discussion.

\section{Conclusion}
Summarize findings and suggest next steps.

\end{document}
```

4.3 Tips

- Use `\tableofcontents` near the beginning for longer documents.
- Keep heading titles short and descriptive.
- Maintain a consistent structure across the document (e.g., each section begins with a short introductory sentence).

5 Including Packages

Packages extend the functionality of LaTeX by adding new commands, environments, and formatting options. Almost every nontrivial document relies on packages for mathematics, figures, tables, hyperlinks, and layout control.

5.1 How to Include a Package

Packages are loaded in the *preamble* of your document (before `\begin{document}`) using the `\usepackage` command:

```
\usepackage{amsmath}
\usepackage{graphicx}
\usepackage{hyperref}
```

5.2 Commonly Used Packages

Here are some of the most common packages you will use:

- `amsmath` — advanced math environments (align, cases, etc.)
- `amssymb` — additional math symbols
- `graphicx` — include images
- `geometry` — control margins and page layout
- `booktabs` — professional-looking tables
- `hyperref` — clickable links and references
- `enumitem` — customize lists
- `xcolor` — colors for text and tables
- `listings` — code listings

5.3 A Minimal Working Example (MWE)

The following example shows how packages are added and used:

```
\documentclass[11pt]{article}

\usepackage{amsmath}
\usepackage{amssymb}
\usepackage{graphicx}
\usepackage{geometry}
\usepackage{hyperref}

```

```latex
\geometry{margin=1in}

\begin{document}

\section{Example}
Here is a clickable link: \url{https://www.overleaf.com}

Here is a matrix:
\[
A =
\begin{bmatrix}
1 & 2 \\
3 & 4
\end{bmatrix}.
\]

Here is a figure:
\begin{figure}[h]
\centering
\includegraphics[width=0.5\linewidth]{example.png}
\caption{An example image}
\end{figure}

\end{document}
```

5.4 Tips

- Always load packages before `\begin{document}`.
- If you get a "command undefined" error, you are missing a required package.
- Overleaf will usually suggest which package you need when a command is missing.
- Avoid loading unnecessary packages to keep compilation fast and stable.

6 Lists (Itemize and Enumerate)

Lists are used to organize information clearly and improve readability. LaTeX provides two main list environments: `itemize` for bulleted lists and `enumerate` for numbered lists.

6.1 Bulleted Lists (`itemize`)

Use `itemize` when the order of items does not matter.

- First item
- Second item

- Third item

6.2 Numbered Lists (`enumerate`)

Use `enumerate` when the order of items is important, such as steps in a procedure.

1. Step one
2. Step two
3. Step three

6.3 Nested Lists

Lists can be nested to represent hierarchical information.

- Main item
  1. Sub-item A
  2. Sub-item B
- Another main item

6.4 Minimal Working Example (MWE)

The following example demonstrates both `itemize` and `enumerate` in a complete document.

```latex
\documentclass{article}
\usepackage{enumitem}

\begin{document}

\section{Experiment Procedure}

\subsection{Materials}
\begin{itemize}
  \item Microcontroller
  \item Sensors
  \item Power supply
\end{itemize}

\subsection{Steps}
\begin{enumerate}
  \item Connect the power supply.
  \item Upload the firmware.
  \item Run the experiment.
  \item Record the data.
\end{enumerate}
```

```latex

\subsection{Notes}
\begin{itemize}
  \item Ensure proper grounding.
  \item Do not exceed voltage limits.
  \begin{enumerate}
    \item Maximum voltage: 12 V
    \item Maximum current: 2 A
  \end{enumerate}
\end{itemize}

\end{document}
```

6.5 Tips

- Use `itemize` for unordered information and `enumerate` for sequences or procedures.
- Avoid overly long list items; break them into multiple sentences if needed.
- Nested lists should be used sparingly to keep documents readable.

7 Writing Mathematics

One of the main advantages of LaTeX is its powerful and professional-quality mathematical typesetting. This section introduces the most commonly used tools for writing equations.

7.1 Inline and Display Math

Inline math is used when a mathematical symbol or expression appears inside a sentence. It is enclosed in dollar signs. For example,

```latex
Let $x(t)$ denote the position.
```

produces "Let $x(t)$ denote the position."

Display math is used for equations that should appear on their own line. It is written using `\[ ... \]`. For example,

```latex
\[
    \dot{x}(t) = A x(t) + B u(t).
\]
```

produces

$$\dot{x}(t) = Ax(t) + Bu(t).$$

7.2 Numbered Equations

To number equations so that you can refer to them later, use the `equation` environment. For example,

```
\begin{equation}
    \dot{x}(t) = A x(t) + B u(t).
    \label{eq:state_space_equation}
\end{equation}
```

produces

$$\dot{x}(t) = Ax(t) + Bu(t). \tag{1}$$

To label and reference equations, use `\label` and `\eqref`. For example,

```
The system dynamics are given by Equation \eqref{eq:state_space_equation}.
```

produces "The system dynamics are given by Equation (1)."

7.3 Aligning Multi-Line Equations

For multi-line equations, use the `align` environment. The `&` symbol indicates where equations should be aligned. For example,

```
\begin{align}
    \dot{x}(t) &= A x(t) + B u(t), \\
    y(t) &= C x(t) + D u(t).
\end{align}
```

$$\begin{aligned} \dot{x}(t) &= Ax(t) + Bu(t), && (2)\\ y(t) &= Cx(t) + Du(t). && (3) \end{aligned}$$

This is especially useful for step-by-step derivations. For example,

```
\begin{align}
    x(t+1) &= A x(t) + B u(t), \\
           &= A (A x(t-1) + B u(t-1)) + B u(t).
\end{align}
```

produces

$$\begin{aligned} x(t+1) &= Ax(t) + Bu(t), && (4)\\ &= A(Ax(t-1) + Bu(t-1)) + Bu(t). && (5) \end{aligned}$$

7.4 Column Vectors

Column vectors are written using the `bmatrix` environment with each entry separated by a line break (`\\`). For example,

```
\[
x =
\begin{bmatrix}
x_1 \\ x_2
\end{bmatrix}.
\]
```

produces

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

7.5 Row Vectors

Row vectors are written using the same `bmatrix` environment, but entries are separated by ampersands (`&`). For example,

```
\[
v =
\begin{bmatrix}
v_1 & v_2 & v_3
\end{bmatrix}.
\]
```

produces

$$v = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}.$$

7.6 Matrices

Matrices use the `bmatrix` environment with rows separated by `\\` and columns separated by `&`. For example,

```
\[
A =
\begin{bmatrix}
0 & 1 \\
-2 & -3
\end{bmatrix}.
\]
```

produces

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}.$$

7.7 Superscripts, subscripts, and symbols

Common LaTeX commands for superscripts, subscripts, and symbols are shown below.

Concept	**LaTeX Code**	**Rendered Output**
Subscript	`x_1`	x_1
Superscript	`x^2`	x^2
Subscript + Superscript	`x_1^2`	x_1^2
Fraction (derivative)	`\frac{dx}{dt}`	$\frac{dx}{dt}$
Fraction (general)	`\frac{a+b}{c+d}`	$\frac{a+b}{c+d}$
Greek letters	`\alpha, \beta, \gamma, \theta, \omega, \lambda`	$\alpha, \beta, \gamma, \theta, \omega, \lambda$
Trigonometric functions	`\sin(x), \cos(x)`	$\sin(x), \cos(x)$
Logarithm and exponential	`\log(x), \exp(x)`	$\log(x), \exp(x)$
Integral and sum	`\int, \sum`	$\int, \sum$

Table 1: Common LaTeX commands for superscripts, subscripts, and symbols.

7.8 Tips

- Do not use screenshots or images for equations; always typeset them.
- Use `align` for derivations instead of forcing line breaks.
- Keep notation consistent throughout your document.

8 Tables

Tables are used to present structured numerical or categorical data in a clear, readable format. In LaTeX, tables are created using the `tabular` environment, which is often wrapped in a `table` environment to support captions and labels. For example,

```
\begin{table}[h]
\centering
    \begin{tabular}{|c|c|c|}
    \hline
    A & B & C \\
    \hline
    1 & 2 & 3 \\
    4 & 5 & 6 \\
    \hline
    \end{tabular}
\caption{Example numerical data}
\label{tab:example}
\end{table}

```

```
Table \ref{tab:example} shows a simple data set.
```

produces

A	B	C
1	2	3
4	5	6

Table 2: Example numerical data

Table 2 shows a simple data set.

For professional documents, tables should include captions and be numbered for referencing. This is done by wrapping `tabular` inside a `table` environment.

8.1 Understanding the Syntax

- The column specification `{|c|c|c|}` means three centered columns separated by vertical lines.
- The symbol `&` separates columns.
- The command `\\` ends a row.
- The command `\hline` draws a horizontal line.

8.2 Tips

- Always include a caption and label for tables you want to reference.

9 Adding Figures

Figures are used to include plots, diagrams, photographs, screenshots, and other visual material. In LaTeX, figures are handled using the `figure` environment together with the `graphicx` package.

9.1 Required Package

To include images, make sure the following line appears in the preamble.

```
\usepackage{graphicx}
```

9.2 Uploading an Image

In Overleaf, upload your image file (e.g., `figure1.png`, `plot.pdf`, or `diagram.jpg`) by dragging it into the file tree on the left or by clicking **Upload**. Use simple file names without spaces.

9.3 Inserting an Image

Once the image is uploaded, you can include it in your document using

```
\begin{figure}[h]
  \centering
  \includegraphics[width=0.75\linewidth]{figure1.png}
  \caption{An example figure.}
  \label{fig:example}
\end{figure}
```

This will place the image in your document, scale it to 75% of the text width, and add a caption and label.

9.4 Understanding the Syntax

- `[h]` requests that the figure be placed "here" (near where it appears in the source).
- `\centering` centers the image.
- `\includegraphics` loads the image file.
- `width=0.75\linewidth` scales the image relative to the text width.
- `\caption{...}` creates a numbered caption.
- `\label{...}` assigns a reference key to the figure.

9.5 Referencing Figures

Once a figure has a label, it can be referenced anywhere in the text.

```
Figure \ref{fig:example} shows the result.
```

This ensures the correct number is always used, even if figures are reordered.

9.6 Multiple Figures and Formats

You can include images in common formats such as.

- `.png`, `.jpg` for photos and screenshots
- `.pdf`, `.eps` for plots and vector graphics

9.7 A Minimal Working Example (MWE)

```
\documentclass{article}
\usepackage{graphicx}

\begin{document}

\section{Results}
Figure \ref{fig:test} shows a simple example.

\begin{figure}[h]
  \centering
  \includegraphics[width=0.6\linewidth]{figure1.png}
  \caption{Sample figure included in a LaTeX document.}
  \label{fig:test}
\end{figure}

\end{document}
```

9.8 Tips

- Always include a caption and label for figures.
- Do not paste screenshots of figures; include the original image files.
- Use consistent figure naming (e.g., `fig1.eps`, `trajectory.png`).

10 Code Listings (Using the `listings` Package)

When writing technical documents, it is often necessary to include source code (e.g., MATLAB, Python, C++). The `listings` package allows you to display code with monospaced formatting, optional syntax highlighting, line numbers, and frames. You will need to add the following packages in the preamble.

```
\usepackage{listings}
\usepackage{xcolor}
```

To improve readability and ensures consistent formatting across your document, you can define a default style for all code blocks using `\lstset`. Add `lstset` right after the usepackage command.

```
\lstset{
  basicstyle=\ttfamily\small,
  numbers=left,
  numberstyle=\tiny,
  frame=single,
```

```
  breaklines=true,
  keywordstyle=\color{blue},
  commentstyle=\color{gray},
  stringstyle=\color{red}
}
```

10.1 Minimal Working Example (MWE)

Use the `lstlisting` environment to include inline code blocks: This will format the code, apply syntax highlighting, and add an optional caption.

```
\begin{lstlisting}[language=Matlab, caption={Example MATLAB snippet for
state-space simulation.}]
m = 1.0; c = 0.6; k = 4.0;
A = [0 1; -k/m -c/m];
B = [0; 1/m];
C = [1 0];
D = 0;
sys = ss(A,B,C,D);

t = linspace(0,10,1000);
u = ones(size(t));     % step force
y = lsim(sys,u,t);
plot(t,y), grid on
xlabel('t(s)'), ylabel('x(m)')
\end{lstlisting}
```

produces

Listing 1: Example MATLAB snippet for state-space simulation.

```
m = 1.0; c = 0.6; k = 4.0;
A = [0 1; -k/m -c/m];
B = [0; 1/m];
C = [1 0];
D = 0;
sys = ss(A,B,C,D);

t = linspace(0,10,1000);
u = ones(size(t));     % step force
y = lsim(sys,u,t);
plot(t,y), grid on
xlabel('t(s)'), ylabel('x(m)')
```

Here is another example.

```
\begin{lstlisting}[language=Python, caption={Example Python Code}]
import numpy as np
```

```
def square(x):
    return x**2
\end{lstlisting}
```

produces

Listing 2: Example Python Code

```python
import numpy as np

def square(x):
    return x**2
```

To include a separate script file (e.g., MATLAB or Python), use

```
\lstinputlisting[language=Matlab]{script.m}
```

10.2 Tips

- Use `listings` instead of screenshots; code remains searchable and clean.
- Keep formatting consistent across assignments and reports.
- Avoid overly long code blocks; move large programs to external files.
- If syntax highlighting fails, check that the correct `language=...` option is set.

11 Citations and References (BibTeX)

Professional technical documents require proper citation of prior work. LaTeX uses BibTeX to manage references in a clean, consistent, and automated way. Overleaf provides built-in support for BibTeX, making it easy to create, edit, and compile bibliographies.

11.1 What is BibTeX?

BibTeX stores all of your references in a separate file (usually `references.bib`). Each reference is given a unique key (e.g., `lamport1994latex`), which you use in your document when citing. This approach allows you to

- Reuse the same reference in multiple places
- Automatically format references in different styles
- Avoid manually typing bibliographies

11.2 Creating a `references.bib` File

In Overleaf, click **New File** → **Blank File**, and name it `references.bib`. Add BibTeX entries like the following:

```
@book{lamport1994latex,
  title = {LaTeX: A Document Preparation System},
  author = {Lamport, Leslie},
  year = {1994},
  publisher = {Addison-Wesley}
}
```

Each entry begins with a type (e.g., `@book`, `@article`, `@inproceedings`) and includes fields such as `author`, `title`, `year`, and `journal` or `publisher`.

11.3 Citing References in the Text

To cite a reference in your document, use the `\cite` command with the citation key.

```
LaTeX was introduced by Lamport \cite{lamport1994latex}.
```

This will automatically insert a citation number or author–year label, depending on the bibliography style.

11.4 Printing the Bibliography

At the end of your document, add:

```
\bibliographystyle{plain}
\bibliography{references}
```

This tells LaTeX to format the bibliography using the `plain` style and to load all entries from `references.bib` that were cited in the document.

11.5 Minimal Working Example (MWE)

```
\documentclass{article}

\begin{document}

This document cites a classic LaTeX reference \cite{lamport1994latex}.

\bibliographystyle{plain}
\bibliography{references}

\end{document}
```

11.6 Common Reference Types

Some common BibTeX entry types include:

- `@article` – journal articles
- `@book` – books
- `@inproceedings` – conference papers
- `@techreport` – technical reports
- `@phdthesis`, `@mastersthesis`

11.7 Tips

- Use Google Scholar, IEEE Xplore, or publishers' websites to export BibTeX entries.
- Keep all references in a single `.bib` file for consistency.
- If a citation does not appear, make sure you ran "Recompile" twice in Overleaf.
- Do not manually type bibliographies when using BibTeX.

12 Citations and References (BibTeX)

Overleaf supports BibTeX for managing references.

12.1 Create a `references.bib` File

Add a new file named `references.bib` and include an entry such as:

```
@book{lamport1994latex,
  title={LaTeX: A Document Preparation System},
  author={Lamport, Leslie},
  year={1994},
  publisher={Addison-Wesley}
}
```

12.2 Cite and Print the Bibliography

In your LaTeX file:

```
Citations look like this \cite{lamport1994latex}.

\bibliographystyle{plain}
\bibliography{references}
```

produces

Citations look like this [1].

References

[1] Leslie Lamport. *LaTeX: A Document Preparation System*. Addison-Wesley, 1994.

13 Common Overleaf Tips

- **Compile errors:** Read the error log; the first error is usually the root cause.
- **Missing packages:** Add the required `\usepackage{...}` in the preamble.
- **File names:** Use simple names without spaces (e.g., `my_plot.pdf`).
- **Version history:** Use Overleaf's history to recover prior versions.

14 Exporting and Submitting

To download your PDF in Overleaf, click **Download PDF**. If a submission requires source files, download the entire project as a ZIP using **Menu** → **Download**.